

Isolation of single somas from postmortem fresh frozen human brain tissue and immunostaining for AT8 and MAP2

Cobos Lab, October 2019

Buffer preparation (All solutions should be RNase-free for single-soma RNAseq experiments):

Isolation Medium #1 (IM1), 45 ml

Prepare in a 50 ml Falcon tube and store at 4°C up to 6 months.

7500 µl	1.5M Sucrose	(final concentration 250mM)
1125 µl	1M KCl	(25mM)
225 µl	1M MgCl ₂	(5mM)
450 µl	1M Tris pH8	(10mM)
35.7 ml	H ₂ O	

Homogenization Buffer (3 ml per sample).

Prepare FRESH and keep ICED or at 4°C. Discard after use.

2955 µl	IM1	
3 µl	DTT 1mM	(final concentration 1µM)
30 µl	50x Protease Inhibitor	(0.5x)
15 µl	RNaseIn 40U/µl	(0.2U/µl)

No Triton!

Iodixanol dilutions

Prepare in a 50 ml Falcon tubes and store at 4°C up to 6 months. Accuracy with Iodixanol and sucrose concentrations is critical.

Iodixanol medium (IDM), 45 ml

7500 µl	1.5M Sucrose	(final concentration 250mM)
3750 µl	1M KCl	(83mM)
750 µl	1M MgCl ₂	(17mM)
1500 µl	1M Tris pH8	(33mM)
31.5 ml	H ₂ O	

Iodixanol 42%(v/v), 20 ml

14 ml Iodixanol 60%(v/v) + 6 ml IDM

Iodixanol 25%(v/v), 30 ml

12.5 ml Iodixanol 60%(v/v) + 17.5 ml IDM

Freezing Storage Buffer (FSB), 15ml

Prepare in a 50 ml Falcon tube and store at 4°C up to 6 months.

1.665 ml	1.5M Sucrose	(final concentration 166.5mM)
75 µl	1M MgCl ₂	(5mM)
150 µl	1M Tris pH8	(10mM)
13.1 ml	H ₂ O	

Buffer for Immunostaining, 10ml

9400 µl	RNase-free PBS pH7.4	
500 µl	RNase-free 10% BSA	(final concentration 0.5%)
50 µl	1M MgCl ₂	(5mM)
10 µl	DNase I (2000U/ml)	(2U/ml)

Buffer for Antibody Incubation

Add to 1ml of Buffer for Immunostaining:

5 µl	RNaseIn 40U/µl	(0.2U/µl)
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Collection medium for FACS (1ml; 0.2 ml/vial).

950 µl	RNase-free PBS pH7.4	
25ul	RNaseIn	
*	BSA 10% *After collecting!	(final concentration 1%)

Reagents

UltraPure™ distilled water, DNase/RNase-free (Thermo Fisher, 10977023)

Sucrose (Thermo Fisher, S25590B)

KCl (Sigma, 60142)

MgCl₂ (Sigma, M2670)

UltraPure™ 1M Tris-HCl Buffer, pH 8.0 (Thermo Fisher, T2944-1L)

DTT (Sigma, D9779-1G)

50x Protease Inhibitor (Roche, 11697498001)

RNaseIn (Lucigen, 30281-2)

DNase I (NEB, M0303S)

BSA (Sigma, A7906)

Iodixanol 60% (v/v) (OptiPrep™ Density Gradient Medium; Sigma, D1556)

Protocol

- Aims for minimal mechanical damage (as little centrifugation and pipetting as possible)
 - Material and tools needed: Forceps, spatula, blades, dounce tissue grinder (Potter-Elvehjem tissue grinder), cell strainers, petri dishes
 - All solutions and materials should be RNase free and kept iced or at 4 °C at all times
 - Glassware and metal tools are sealed with aluminum foil and baked at 220 °C for 6 h
1. Prepare Homogenization Buffer and cool it on ice.
Pre-cool the dounce tissue grinder on ice
(**Potter-Elvehjem tissue grinder**, 8 mL tubes, 0.1–0.15 mm clearance space between pestle and tubes (Kimble, K886000-0022))
Add 2.4 ml of Homogenization buffer to the dounce tissue grinder.
 2. Collect the brain chunk (~200 mg) and transfer to a Petri dish on ice.
Cut out into small pieces using a chilled scalpel or blade.
 3. Transfer all pieces into the dounce tissue grinder. Slowly homogenize the tissue, on ice (**8~15 strokes; slow**). Check on hemacytometer while homogenizing and adjust the number of strokes (too few strokes – you will see clumps; too many strokes – you will get a higher proportion of naked nuclei).
 4. Filter homogenate using a **100 µm** Corning cell strainer to remove clumps.
→Take sample for hemocytometer.
 5. Transfer the homogenate into two precooled 1.5 ml Eppendorf tubes.
Centrifuge at **400g for 5 min** at 4°C.
 6. Slowly aspirate the supernatant from the side of the tube. Avoid disturbing the pellets, up to 50 µl of supernatant can be left.
Gently **resuspend** each pellet in 225 µl (final volume) of cold Homogenization buffer and pool both tubes (final volume 450 µl).
 7. Add an **equal** volume of cold **42%v/v iodixanol**. (Critical! Be very precise with volumes. 450 µl suspension + 450 µl 50% iodixanol; Final iodixanol concentration is 21%). Gently pipette mix.
 8. Add an equal volume of **25% iodixanol** (900 µl) into a 2 ml Eppendorf tube. Then, slowly layer off the 21% iodixanol suspension mix over the 25% iodixanol, without mixing them.
Centrifuge at **8,000g for 15 min** at 4°C.

→Prepare buffer for Immunostaining
 9. After centrifugation, remove and discard the top myelin-rich debris layer (you can use a 1 ml pipette with the tip cut, or cotton swabs). Then, remove and discard the aqueous supernatant, without disrupting the **somas pellet**. Avoid contaminating with the top layer.
Use a small amount of **buffer for Antibody Incubation** to resuspend the pellet and transfer the solution to a new tube.

10. Gently resuspend in 200 µl of **buffer for Antibody Incubation**. Incubate for 15 min, at 4°C or iced, for **blocking** nonspecific staining.
→ Take sample for hemocytometer.
11. Take a sample for **unstained** control. Take sample for **2AB**-only (nonspecific binding control).
12. Add primary antibodies (**Ms-a-AT8 1:150 and Rb-a-MAP2 1:40**) and incubate on a shaker in cold room (4°C) for 40 min.
(MAP2 antibody: Millipore #AB5622, p-Tau (AT8) antibody: Thermo Fisher #MN1020)

→ Note: use Eppendorfs coated with BSA for collection (to coat the Eppendorf tubes, fill them with 10% BSA solution in PBS for 5 min, rinse with PBS, and dry at 4°C overnight).
13. **Washing**. Add 500 µl of buffer for Immunostaining and invert several times. Centrifuge at 400g for 5 min at 4 °C. Carefully transfer all the supernatant (don't leave any supernatant! It's ok to drag some pellet) to a new Eppendorf tube and resuspend the pellet in 400 µl of buffer for Antibody Incubation. Centrifuge the supernatant at 500g for 5 min at 4 °C to recover non-pelleted somas, resuspend them in 200 µl of buffer for Antibody Incubation and pool for a final volume of 600 µl.
14. Take samples for single color staining.
15. Add secondary antibodies (**G-a-Ms Alexa350 1:500 and G-a-Rb Alexa647 1:500**) and **Sytox Green 1:400** (1:100 stock, final concentration **1:40.000**)

→ **Check** the quality of the sample on hemocytometer. For 200 mg of tissue, we should have ~ 1 Million somas in ~0.6 mL.
16. Transfer to 7ml culture tubes. Place on ice and bring them to the FACS facility for sorting.
17. Collect in BSA-coated Eppendorfs containing collection medium (volume of collection medium should be ~1/5 of the expected final volume after collection).
Add BSA after collection for 1% BSA final concentration (tested for 10x Chromium v2 and v3). Different Single Cell assays tolerate different BSA concentrations, but lowering it may increase nuclei aggregation.

FACS sorting controls:

Non-stained

2AB only

Single stainings